

MIT653 RESEARCH METHODS | ASSESSMENT 2-2: LITERATURE REVIEW PRESENTATION

The Effectiveness of Generative AI in Creating Personalised Video and Audio Content

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Three core research questions evaluate quality, efficiency, and ethics of GenAI

RQ1: QUALITY

What effects does generative AI have on the relevance, quality, realism, and inventiveness of customized audio and video content?

Focus Areas:

- Visual fidelity and semantic alignment
- Temporal/physical consistency
- Voice and audio naturalness

RQ2: EFFICIENCY

How can generative AI enhance digital content production's scalability, accessibility, and efficiency?

Focus Areas:

- Automating production steps
- Cost-effective scalability
- True efficiency gains vs. review bottleneck

RQ3: ETHICS

What ethical issues are raised by AI-generated audio and video in terms of authenticity, copyright, privacy, and bias?

Focus Areas:

- Deepfakes and authenticity
- IP and training data copyright
- Voice privacy, cloning, and social bias

Ten high-impact studies from 2020–2026 form our literature review matrix

Rigorous Search Methodology

- Time Frame: 2020 - 2026 (rapid tech leaps)
- Source Types: Peer-reviewed journals & conference papers
- **Keywords Used:**
 - Generative AI, Text-to-Video, Video Synthesis
 - Voice Cloning, Generative Audio, Speech Synthesis
 - Deepfake, AI Ethics, Copyright, Personalisation

Selected 10 landmark papers for thematic mapping to balance technical models and ethical governance.

Study (Year)	Focus Domain	Key Finding	Relevance
Singer et al. (2022)	Video Synthesis	Make-A-Video text-to-video capabilities	Video Quality
Kondratyuk (2024)	Multimodal AI	VideoPoet zero-shot cross-modality generation	Personalisation
Le et al. (2023)	Speech Generation	Voicebox multilingual universal cloning	Production Efficiency
Barnett (2023)	Audio Ethics	Reviewed 884 papers; huge lack of risk research	Ethics & Risks
Amankwah (2024)	Creative Work	AI disruption vs. productivity boosts	Labour Impact

VideoPoet and Voicebox lead the push toward high-definition multimodal synthesis

Video Quality & Personalisation

- Visual Fidelity Leap: Diffusion models (Imagen Video) and multimodal LLMs (VideoPoet) have dramatically improved semantic alignment and resolution.
- **Core Technical Gaps:**
 - Temporal Inconsistencies: Characters/backgrounds morph across frames.
 - Physical Anomalies: Object interaction violates real-world physics.
- Evaluation Gap: Traditional automated metrics fail to capture subjective user satisfaction, relevance, and cultural appropriateness.

Audio & Speech Personalisation

- Sound Synthesis Progress: MusicLM and MusicGen enable precise text-conditioned music generation. Voicebox achieves state-of-the-art multilingual speech synthesis.
- **Major Audio Limitations:**
 - Fine-Grained Controllability: Directing specific vocal beats remains hard.
 - Emotional Depth: Cloned speech often lacks authentic human emotions.
- Personalisation Need: Synthesized content must adapt directly to audience identity, preferences, and cultural background.

Generation speed does not equal true workflow efficiency due to human check bottleneck

TRADITIONAL WORKFLOW (Linear & Resource-Heavy)

Planning & Scripting -> Studio Recording -> Manual Video/Audio Editing -> Post-Production -> Approval

- Constraints: Extremely high human labor, long lead times, expensive, and near-zero scaling potential.

GENAI-ASSISTED WORKFLOW (Instant Generation + Critical Human Oversight)

Text Prompting -> Instant AI Rendering -> HUMAN REVIEW, FACT CHECK, COPYRIGHT & ETHICAL APPROVAL -> Launch

- Leverage: Highly scalable multi-language generation. But the real bottleneck is human-in-the-loop compliance checks.

The True Efficiency Metric:

Do not equate 'generation speed' with 'production efficiency'. Effective use must measure the entire workflow from the first prompt to the final compliant, ethically-approved output.

Severe ethical concerns persist around deepfakes, copyright ownership, and voice privacy

1. Authenticity & Deepfakes

Audiences cannot easily distinguish realistic synthetic media from truth, fueling massive fraud, political misinformation, and reputational damage.

2. Copyright & Intellectual Property

GenAI training relies heavily on scraping existing creative works without credit or compensation, resulting in legal battles over authorship and fair use.

3. Privacy & Consent

Voice cloning allows unauthorized imitation of unique vocal identities, necessitating mandatory consent systems and biometric control structures.

4. Stereotyping & Social Bias

Training datasets often perpetuate and amplify human prejudices, which then inadvertently get baked into personalized local content.

Regulatory Outlook: Emerging frameworks like the EU AI Act (2024) enforce mandatory transparency, watermark labelling, and strict human oversight.

Critical Gap: Less than 10% of generative-audio research addresses negative ethical impacts

< 10%

OF GENERATIVE-AUDIO PAPERS COVERS
NEGATIVE IMPACTS

Key Research Gaps Identified in Literature

- **Technology vs. Impact Imbalance:** Barnett's (2023) systematic review of 884 audio papers revealed a massive development bias. Technical capability advances rapidly, while impact, governance, and mitigation studies are severely neglected.
- **Lack of Real-World Evaluation:** Most model research is tested in sanitized laboratory environments. There is a deficit of empirical studies analyzing GenAI inside actual commercial workflows, audiences' psychological trust, or long-term effects on creative laborers.
- **Fragmented Governance:** Current literature successfully lists risks but fails to evaluate practical, scalable tools like audio provenance, consent verification systems, and training data copyright management.

Human-AI collaboration is the most sustainable path for effective creative work

Generative AI Capabilities

- Scale & Speed: Rapid synthesis of multi-language and high-volume media.
- Technical Accessibility: Lowering entry barriers for non-expert creators.
- Cost-Efficiency: Drastic cuts in video rendering and recording overhead.

Human Creative Oversight

- Contextual Judgment: Adapting tone, cultural relevance, and authentic emotions.
- Quality & Brand Control: Ensuring temporal consistency and physical realism.
- Ethical & Legal Governance: Fact-checking, copyright clearance, and consent.

Core Synthesis: The Balanced Human-AI Collaboration Model

Our review argues that the optimal model is not full automation, but a deeply integrated human-AI partnership.

- Production: AI acts as a raw generator to produce assets at high speed and scale.
- Direction: Humans act as creative directors, quality gatekeepers, and ethical judges to ensure reliability.
- Future Focus: Development of comprehensive evaluation frameworks combining technical metrics and human judgment.